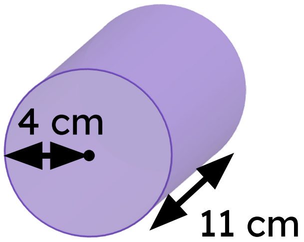
Surface area of cylinders

Task A: Finding the surface area of a cylinder

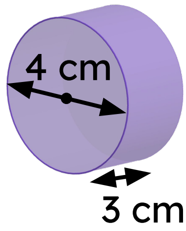
1) Work out the surface area of the following **cylinders**

a)



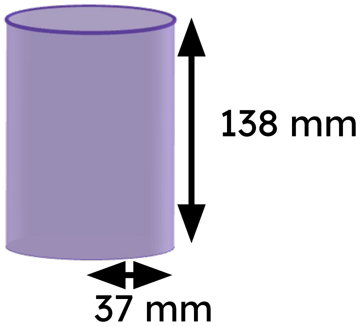
Surface area
$= 2 \times \pi \times 4^2 + 2 \times \pi \times 4 \times 11$
$= 32\pi + 88\pi$
$= 120\pi = 377.0$ cm^2 (1 d.p.)

b)



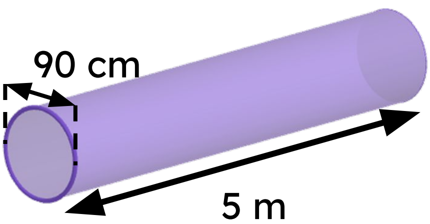
Surface area
$= 2 \times \pi \times 2^2 + \pi \times 4 \times 3$
$= 8\pi + 12\pi$
$= 20\pi = 62.8$ cm^2 (1 d.p.)

c)



Surface area
$= 2 \times \pi \times 37^2 + 2 \times \pi \times 37 \times 138$
$= 2738\pi + 10\ 212\pi$
$= 12\ 950\pi = 40\ 683.6$ mm^2 (1 d.p.)

d)

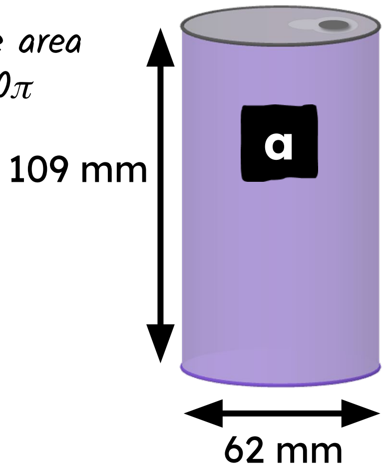


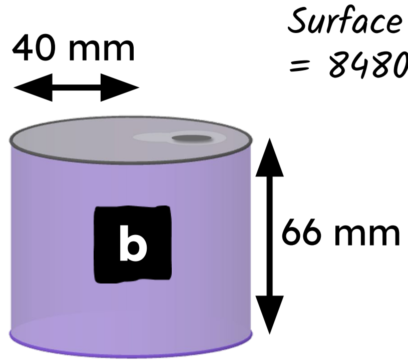
Surface area (working in metres)
$= 2 \times \pi \times 0.45^2 + \pi \times 0.9 \times 5$
$= 0.405\pi + 4.5\pi$
$= 4.905\pi = 15.4$ m^2 (1 d.p.)

2) A drinks company are looking to use less material when making their cans. These two cans both hold 330 ml of liquid. Which can should the drinks company choose?

Drinks company should choose can b.

Surface area $= 8680\pi$





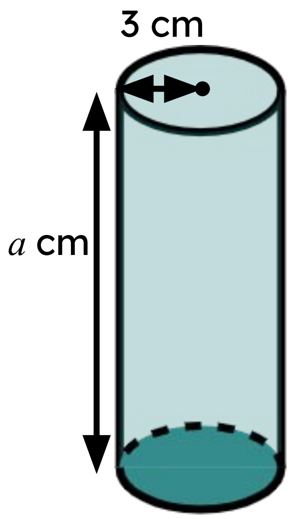
Surface area $= 8480\pi$

Surface area
$= 2 \times \pi \times 31^2 + \pi \times 62 \times 109$
$= 1922\pi + 6758\pi$
$= 8680\pi = 27\ 269.0\ mm^2$ (1 d.p.)

Surface area
$= 2 \times \pi \times 40^2 + 2 \times \pi \times 40 \times 66$
$= 3200\pi + 5280\pi$
$= 8480\pi = 26\ 640.7\ mm^2$ (1 d.p.)

Task B: Using the surface area of a cylinder

1) Find the missing length in each question. Give your answer in cm.



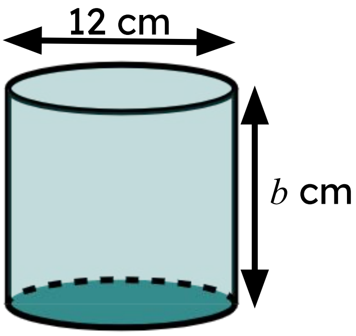
Surface area
$= 78\pi$ cm^2

$2 \times \pi \times 3^2 + 2 \times \pi \times 3 \times a = 78\pi$

$18\pi + 6\pi a = 78\pi$

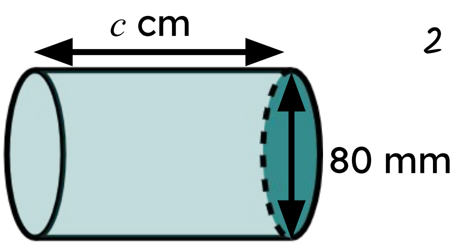
$6\pi a = 60\pi$

$a = 10$ cm



Surface area
= 168π cm^2

$$2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times b = 168\pi$$
$$72\pi + 12\pi b = 168\pi$$
$$12\pi b = 96\pi$$
$$b = 8 \text{ cm}$$



Surface area
= 88π cm^2

$$2 \times \pi \times 4^2 + 2 \times \pi \times 4 \times c = 88\pi$$
$$32\pi + 8\pi c = 88\pi$$
$$8\pi c = 56\pi$$
$$c = 7 \text{ cm}$$

2) A necklace is being made. Part of the design is a metal cylindrical rod.

The rod is gold-plated. 208π mm^2 of gold leaf is used. The rod has a diameter of 4 mm.

How long is the rod?

$$2 \times \pi \times 2^2 + 2 \times \pi \times 2 \times l = 208\pi$$
$$8\pi + 4\pi l = 208\pi$$
$$4\pi l = 200\pi$$
$$l = 50 \text{ mm}$$